KP22 Robust Statistical Regression and Ridge Regression

> Level: undergraduate, year 3
>
> Pattern Recognition | Chapter 4, §4.19.2

Slides: read and download

Learning objectives

- Explain why least squares fears outliers and write the Huber and ε-insensitive losses.
- Explain how ε-insensitive regression produces sparsity and the trade-off in ε.
- Write the closed form and the two forms of ridge regression and compare what each of the four methods resists.

KP22 Robust Statistical Regression and Ridge Regression

One-sentence summary: Least squares is terrified of outliers; with a loss that is lenient on large errors, one bad point no longer bends the line. Huber makes large errors linear, the ε-insensitive loss ignores small errors (and gains sparsity) and ridge regression takes ε=0 to gain a closed form.

1. The outlier

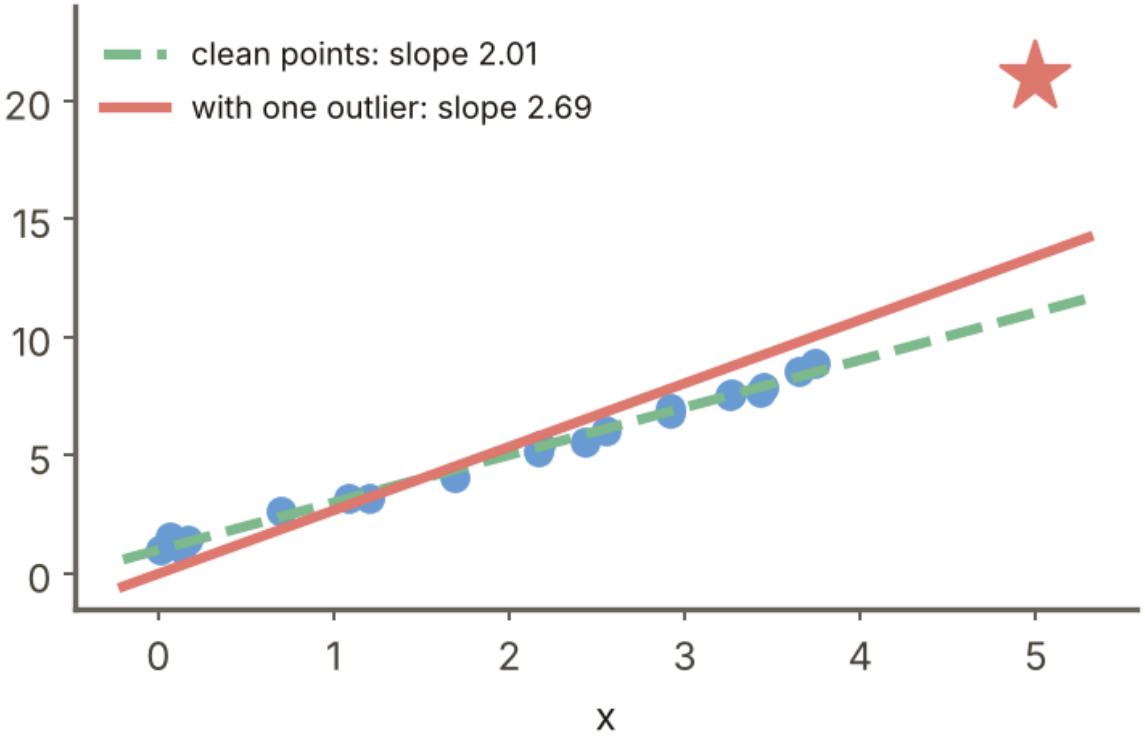
Plain explanation: What does least squares do when one point is badly wrong?

- **Outlier**: a point far from the trend of the others (an extreme value or a wrong label); the effect is larger at the edge of the x range.

Example (designed here): 20 points $y = 2x + 1 +$ noise (sd 0.3) with $x \in [0, 4]$, plus one outlier at $x = 5$ with y raised by 10. The least-squares slope goes $2.009 \rightarrow$

2.686 and the intercept $1.000 \to -0.007$ (truth 2 and 1); three noise draws give 2.69, 2.95 and 2.98. Squaring makes a large error's contribution grow quadratically, so it dominates the solution.

Figure `lec05-kp22-outlier`: 20 points, the outlier (star) and two least-squares lines.



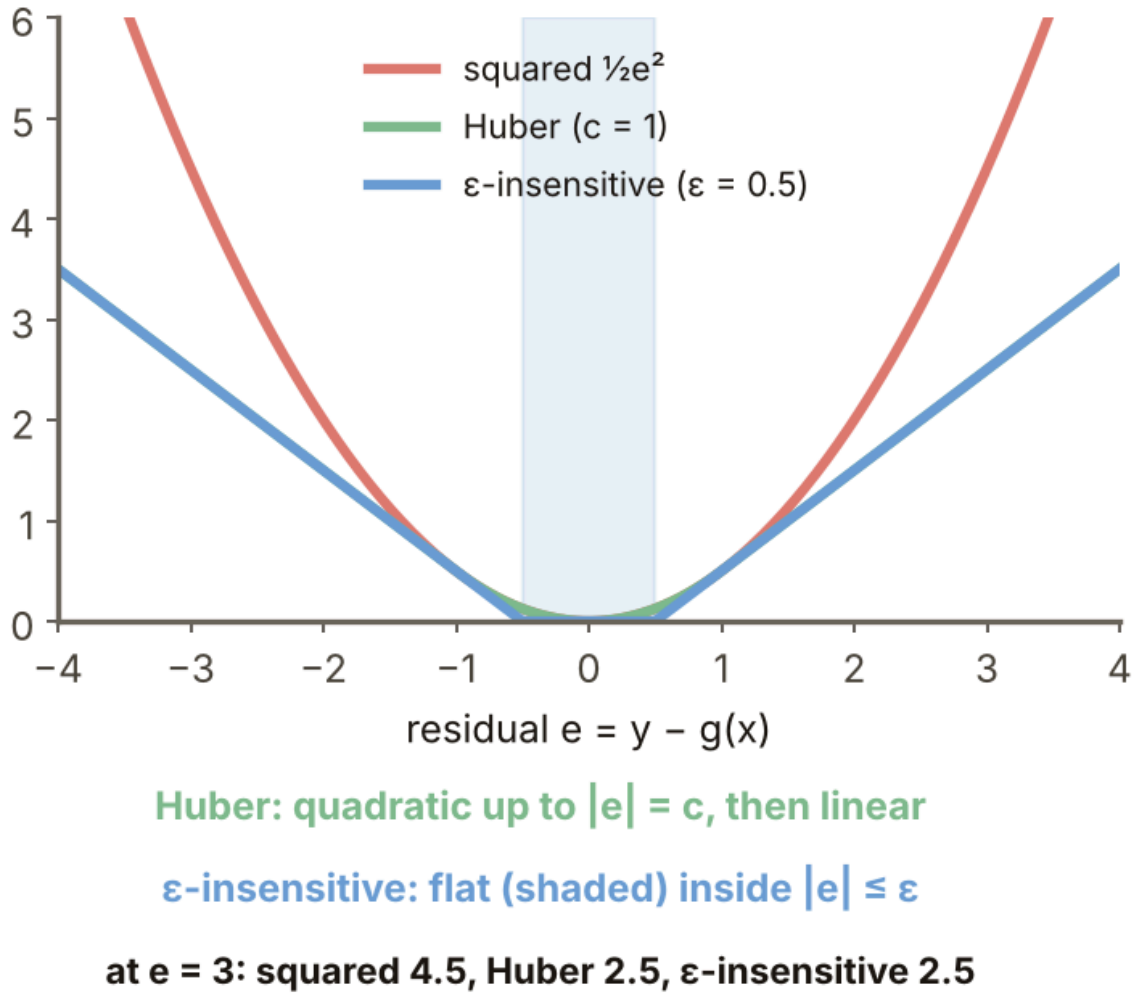
2. Robust losses

A **robust loss** is lenient on large errors so that a few bad points no longer dominate. The **ε-insensitive loss** is $\max(0, |e| - \epsilon)$; errors within ϵ cost nothing. The **Huber loss**:

$$\mathcal{L} = \begin{cases} \frac{1}{2}e^2 & |e| \le c \\ c|e| - \frac{c^2}{2} & |e| > c \end{cases}$$

The outlier's influence drops from quadratic to linear. Values: $e = 3$: squared 4.5, Huber ($c = 1$) 2.5, ε-insensitive ($\epsilon = 0.5$) 2.5; $e = 10$: 50, 9.5, 9.5.

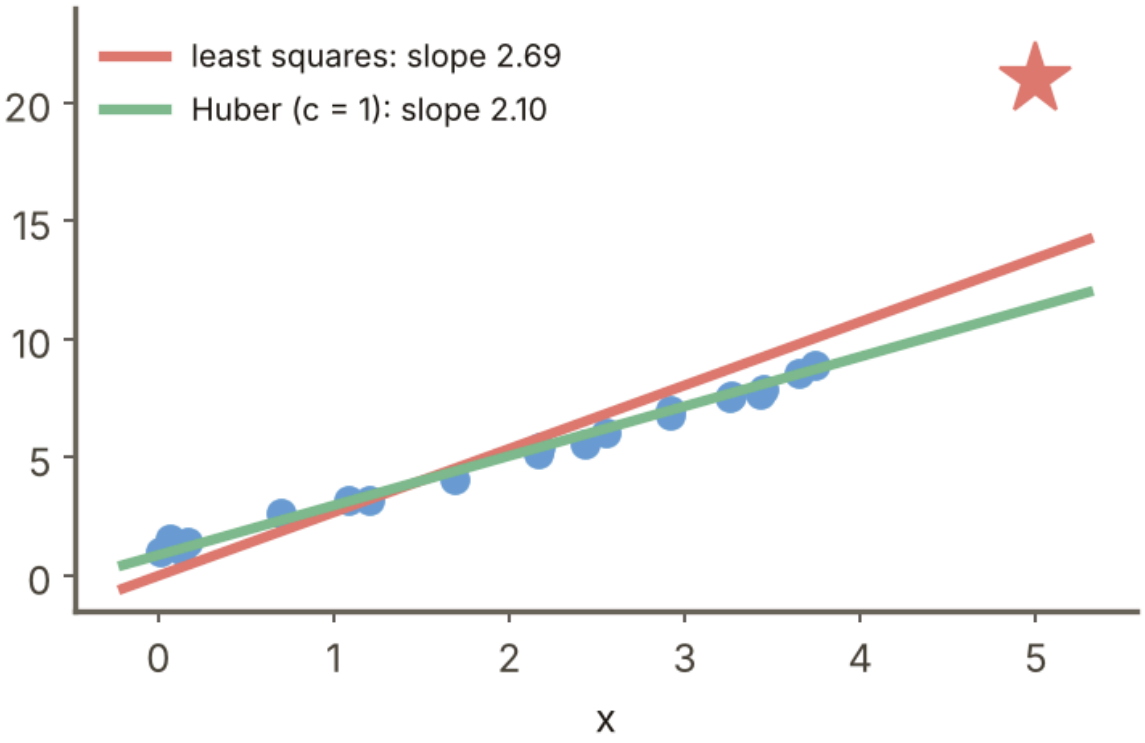
Figure `lec05-kp22-losses`: three losses against the residual.



3. Huber fit

IRLS (iteratively reweighted least squares): weight 1 when $|e| \le c$ and $c/|e|$ otherwise, repeated until convergence. **Example** (designed here, the same 21 points): Huber ($c = 1$) slope 2.099, intercept 0.867; least squares 2.686, -0.007; least squares on clean data 2.009, 1.000. Three noise draws: Huber 2.10, 2.14, 2.13; least squares 2.69, 2.95, 2.98. Huber is still slightly off (truth 2) but far less.

Figure `lec05-kp22-hfit`: the two lines on the same 21 points.



same 21 points, true slope 2 and intercept 1

Huber: slope 2.10, intercept 0.87

least squares: slope 2.69, intercept -0.01

4. ε-insensitive regression

Formal definition (SVR): minimize $\frac{1}{2}\|w\|^2 + C\sum(\xi_i + \xi_i^\star)$ with $|y_i - \langle w, \phi(x_i)\rangle - w_0| \leq \epsilon +$ slack; $g(x) = \sum_i(\lambda_i^\star - \lambda_i)K(x, x_i) + w_0$. Read from KKT: error $< \epsilon \rightarrow \lambda_i = \lambda_i^\star = 0$; support vectors = points with error $\geq \epsilon$; error $> \epsilon \rightarrow \lambda = C$. ϵ is the sparsity knob.

Figure `lec05-kp22-svr`: the objective, the regressor and the KKT readings.

$$\text{minimize } \tfrac{1}{2}\|w\|^2 + C \sum (\xi_i + \xi_i^*)$$

tube: $|y_i - \langle w, \varphi(x_i)\rangle - w_0| \le \varepsilon + \text{slack}$

$$g(x) = \sum_i (\lambda_i^* - \lambda_i)\, K(x, x_i) + w_0$$

KKT readings

$|\text{error}| < \varepsilon \;\rightarrow\; \lambda_i = \lambda_i^* = 0$ **(no contribution)**

support vectors: $|\text{error}| \ge \varepsilon$

$|\text{error}| > \varepsilon \;\rightarrow\; \lambda_i = C$ **(at the upper bound)**

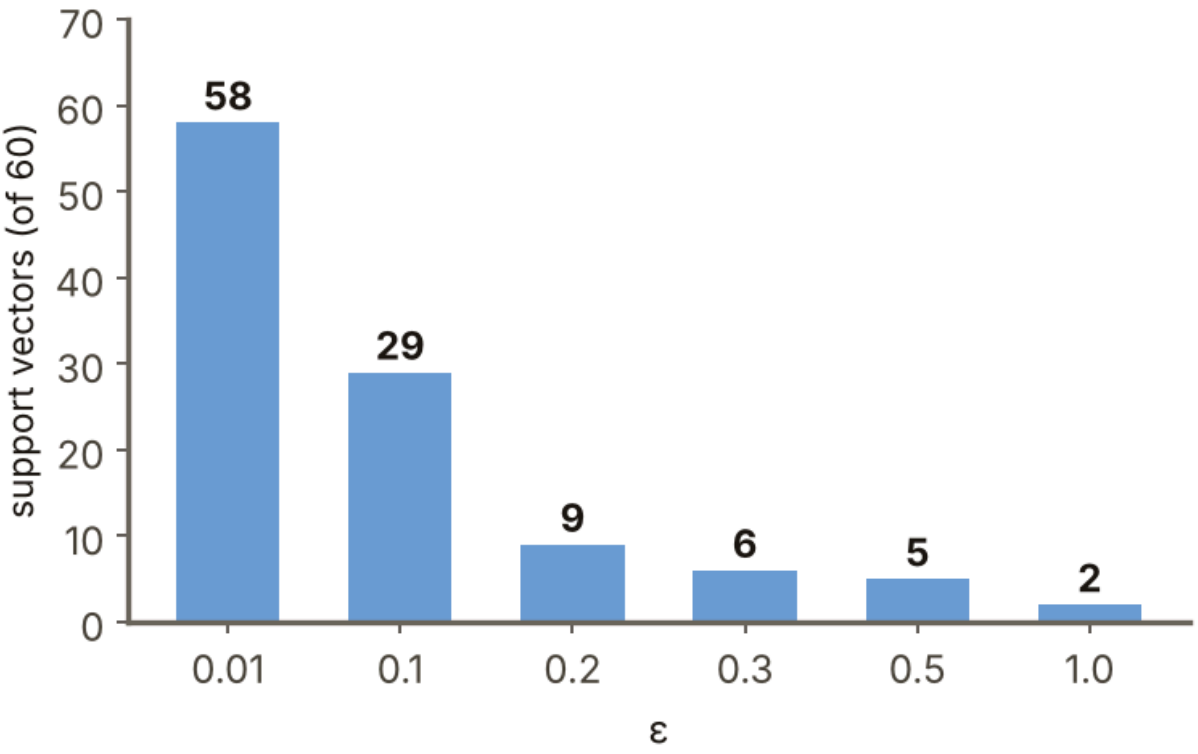
ε is the sparsity knob

(support vectors: the points on or outside the tube)

5. Turning ε

Example (designed here): $y = \sin x +$ noise (sd 0.15), 60 points, Gaussian kernel $\gamma = 0.5$, $C = 10$. Support vectors: $\epsilon = 0.01$: 58; 0.1: 29; 0.2: 9; 0.3: 6; 0.5: 5; 1.0: 2. Mean squared error against the true sin: 0.0008, 0.0020, 0.0038, 0.0175, 0.0551, 0.3455. A larger ϵ gives a sparser but coarser fit.

Figure `lec05-kp22-eps`: support vectors against ϵ.



error against sin: 0.0008 (ε = 0.01) → 0.0551 (ε = 0.5)

a larger ε keeps fewer support vectors

but the fit gets worse: sparsity is paid for

6. Ridge regression

Ridge regression: least squares with a penalty on the norm, done in the RKHS. Set $\epsilon = 0$, $w_0 = 0$ and use equalities: minimize $\mathcal{C}\|w\|^2 + \sum \xi_i^2$ with $\mathcal{C} = \frac{1}{2C}$. The closed form is $g(x) = y^T(\mathcal{K} + \mathcal{C}I)^{-1}p$; it differs from the kernel least squares of KP21 only by $\mathcal{C}I$. The price: with $\epsilon = 0$ no point is a free zero-error point, so sparsity is lost (the loss became squared and the constraints equalities); there is no free lunch.

Figure `lec05-kp22-ridge`: the objective, the closed form and the two linear-kernel forms.

ε = 0, w_0 = 0, equality constraints:

minimize $C\|w\|^2 + \Sigma\, \xi_i^2$, $\xi_i = y_i - \langle w, \varphi(x_i)\rangle$

$$g(x) = y^T (K + CI)^{-1} p$$

linear kernel, two forms of the same w:

$$w = (X^T X + CI)^{-1} X^T y$$

$$w = X^T (XX^T + CI)^{-1} y$$

15 random points, 4 features, C = 0.8: gap 6e-16

no ε, so no zero coefficients

7. Collinearity

The two linear-kernel forms: $w = (X^T X + \mathcal{C} I)^{-1} X^T y = X^T (X X^T + \mathcal{C} I)^{-1} y$; with 15 points, 4 features and $\mathcal{C} = 0.8$ the largest gap is 6×10^{-16}. **Example** (designed here): two nearly identical columns (differing by 10^{-4}): $\mathcal{C} = 0$: $w = (-170.0, 171.0)$, condition number 5×10^8; $\mathcal{C} = 0.1$: $(0.49, 0.49)$, condition number 776. Adding a small positive number on the diagonal is a standard numerical stabilizer; the price is coefficients pulled toward 0.

Figure `lec05-kp22-collinear`: coefficients and condition numbers for different $\mathcal{C}$.

C	w_1, w_2	cond(X^TX + CI)
0	-169.98, 170.96	5.14e+08
0.001	0.47, 0.52	7.75e+04
0.1	0.49, 0.49	776
1	0.49, 0.49	78.5

two almost identical columns: both y ≈ x_1 and y ≈ x_2 fit

C = 0 gives −170 and 171; a small C gives about 0.5 and 0.5

8. Four methods compared

On the same 21 points with the outlier, slopes (truth 2): least squares 2.686; Huber ($c = 1$) 2.099; ε-insensitive ($\epsilon = 0.5$, $C = 10$) 2.122; ridge ($\mathcal{C} = 0.1$) 2.679. Nonzero coefficients: ε-insensitive has only 4 support vectors; the other three use all 21. **To remember**: Huber = large errors linear; ε-insensitive = small errors ignored (sparse); ridge = $\epsilon = 0$ for a closed form (not sparse). **Common misconception**: ridge regression resists outliers; in fact it resists collinearity and overfitting, not outliers.

Figure `lec05-kp22-compare`: slopes and nonzero coefficients of the four methods.

method	slope	nonzero
least squares	2.686	21 of 21
Huber (c = 1)	2.099	21 of 21
ε-insens. (ε = 0.5)	2.122	4 of 21
ridge (C = 0.1)	2.679	21 of 21

true slope 2; the clean-data fit gives 2.009

Huber and the ε loss resist the outlier; ridge does not

only the ε loss gives a sparse solution

Extension: the threshold c, practice and worked solution

The Huber threshold c (the same 21 points): $c = 0.2$: 2.045; 0.5: 2.054; 1: 2.099; 2: 2.188; 5: 2.455; 20: 2.686. A large c puts every error in the quadratic zone, which is least squares; a small c nearly ignores the outlier. Choose c to match the noise level (sd 0.3 here).

Figure `lec05-kp22-csweep`: the Huber slope against c.

slope of the Huber line on the same 21 points

c	slope
0.2	2.045
0.5	2.054
1	2.099
2	2.188
5	2.455
20	2.686

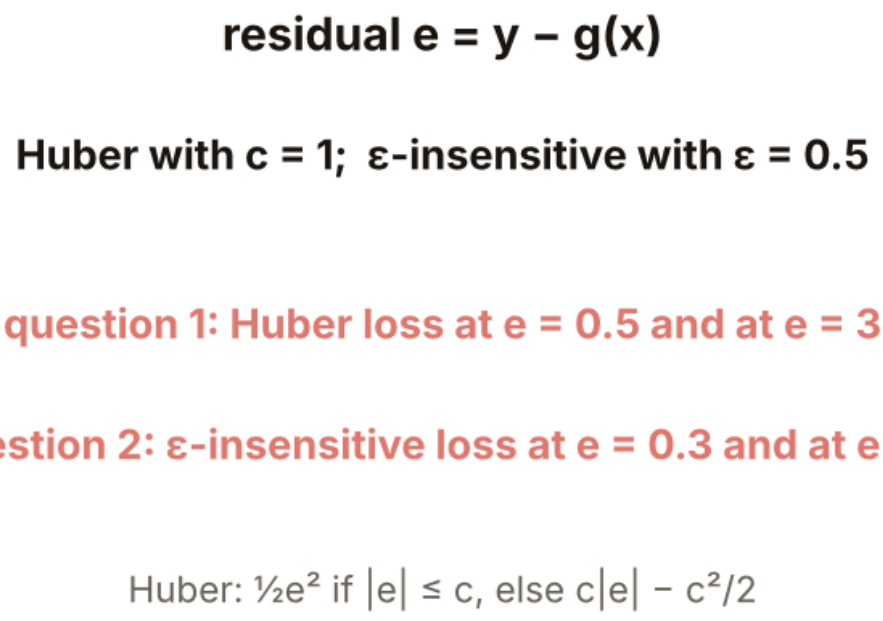
c large: same as least squares (2.686)

c small: nearly ignores the outlier

Practice: residual $e = y - g(x)$; Huber with $c = 1$ and ε-insensitive with $\epsilon = 0.5$.

1. What is the Huber loss at $e = 0.5$ and at $e = 3$?
2. What is the ε-insensitive loss at $e = 0.3$ and at $e = 2$?

Figure `lec05-kp22-practice`: the setup of the practice.



Worked solution of questions 1 and 2: (1) $e = 0.5 \leq 1$: $\frac{1}{2} \cdot 0.25 = 0.125$; $e = 3 > 1$: $1 \cdot 3 - \frac{1}{2} = 2.5$; the squared $\frac{1}{2}e^2$ would give 0.125 and 4.5. (2) $e = 0.3$: $\max(0, 0.3 - 0.5) = 0$; $e = 2$: $\max(0, 2 - 0.5) = 1.5$. (3) Check: a small error costs the same under Huber and squared but 0 under ε-insensitive; a large error grows linearly under both robust losses.

Figure `lec05-kp22-practice-sol`: the computation.

question 1: $e = 0.5 \le 1$, Huber $= \tfrac{1}{2} \cdot 0.25 = 0.125$

$e = 3 > 1$, Huber $= 1 \cdot 3 - 1/2 = 2.5$

(squared $\tfrac{1}{2}e^2$ would give 0.125 and 4.5)

question 2: $e = 0.3$, $\max(0, 0.3 - 0.5) = 0$

$e = 2$, $\max(0, 2 - 0.5) = 1.5$

small errors vanish, large ones grow linearly

Terms and source

中文	English
離群點	Outlier
穩健損失	Robust loss
ε-不敏感損失	ε-insensitive loss
脊迴歸	Ridge regression

Source: Chapter 4, KP22, §4.19.2 (read 2026-10-25). Least squares fearing outliers, the three robust losses, the KKT reading and sparsity of ε-insensitive regression, the closed form and two forms of ridge regression and the 'to remember' line follow the source; the outlier and Huber experiments, the ε sweep, the collinear example, the four-method comparison, the c sweep and the practice were designed and recomputed for this lecture ([added], (program-computed)).